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Quizzes to increase success in mathematics courses for engineering students

HELENA JOHANSSON* AND PETER GLANS

Department of Engineering, Mathematics and Science Education, Mid Sweden University, Holmgatan 10, SE-851 70 Sundsvall, Sweden

**Corresponding author. Email:* helena.johansson@miun.se

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Mathematics is a central component of engineering education, yet many students struggle to succeed in first-year university mathematics courses, resulting in low pass rates and high attrition. Previous research highlights the importance of fostering continuous engagement with mathematical content, but how to achieve this remains a challenge. This study investigates the potential of weekly online quizzes to support engineering students' success in mathematics. Weekly quizzes were introduced in two calculus courses for first-year engineering students at a Swedish university. Quantitative analyses compared exam results in calculus to students' prior performance in linear algebra, while three surveys captured students' perceptions of the quizzes. While effect sizes were moderate, the findings suggest that quizzes can support students' understanding and retention when implemented regularly and in a low-stakes format. Survey data further revealed that most students perceived the quizzes as helpful. However, quizzes were not ranked as the most useful learning resource, with lectures consistently valued more highly, and students indicated that bonus points for completion of quizzes were crucial for their participation. Moreover, the quizzes primarily reached students already engaged in the course. The study concludes that weekly quizzes can promote continuous learning and complement traditional instruction, but they are unlikely to address low pass rates on their own. Their effectiveness depends on thoughtful integration with other teaching practices, as well as strategies to engage students who are less active with their studies.

1. Introduction

Mathematical knowledge is generally agreed to be fundamental for engineering students, and mathematics courses are part of engineering programs all over the world. While studies have shown the importance of mathematics for completion of an engineering degree (Tsui & Khan, 2023), results from various countries show a decline in students' mathematical preparedness when entering engineering programs (Hodgen *et al*., 2018; Nortvedt & Siqveland, 2019), and many countries struggle with high drop-outs in mathematics (Pepin *et al*., 2021). Particularly, engineering programs face low student achievement in

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mathematics and low retention rates, reflected in high dropout rates (e.g. Evans *et al*., 2021), which is also the case in Sweden (Swedish Higher Education Authority, 2023, pp. 31–32).

Previous research has shown that students' attitudes towards university studies and different study strategies affect their success in mathematics courses (e.g. Maciejewski & Merchant, 2016; Laging & Voßkamp, 2017). For example, the common strategy of primarily using textbooks in mathematics courses to find patterns and procedures in examples to memorize and imitate one's own solutions negatively affects students' success in mathematics (Lithner, 2003; Büchele, 2023). In a study focused on the effect of engineering students' study strategies on their success in calculus, factor analyses showed that the strategy of imitating readymade solutions when studying for exams had a negative effect on exam results, while regular learning (study at a consistent pace throughout the course) had a positive effect (Johansson *et al*., 2021). The odds of succeeding on the exam were about 3.4 times higher if students focused on regular learning than if they did not. Meanwhile, the odds of success decreased by about 75% with an imitative learning strategy. Furthermore, students that focus on imitation and memorizing are less likely to engage in regular learning (Büchele, 2023). In particular, regular learning through the use of assignments has been shown to improve mathematics performance (Laging & Voßkamp, 2017). At the same time, other studies have shown varying results regarding the effect of assignments on students' success in different subjects (e.g. Gutarts & Bains, 2010; Hellem & Loras, 2020; Latif & Miles, 2020; Gunawan *et al*., 2022), and some studies have shown that quizzes may even be better for students' results than assignments (Fernandez *et al*., 2006).

In summary, previous research shows that to increase students' chances of succeeding in mathematics courses, and thus improve course completion rates, it seems important to establish a habit among students to continuously engage with the mathematical content throughout the course. How to help students develop this habit of regular learning is not straightforward, and we need to know more about how different ways of encouraging students impact their learning outcomes. Particularly, there is a need to know more about whether quizzes lead to better learning experiences (e.g. Figueroa-Cañas & Sancho-Vinuesa, 2021).

2. Purpose and research questions

The purpose of this study is to contribute to the understanding of ways to improve engineering students' pass rates on mathematics courses. In this study, this is delimited to the use of quizzes, and we will answer the following research questions (RQ):

1) What effect does the regular use of quizzes have on student success in mathematics courses?
2) What are students' perceptions of the use of regular quizzes in mathematics courses?

The term effect is here used to denote observed associations between quiz participation and exam outcomes, rather than causal effects in an experimental sense.

3. Literature review of previous studies on quizzes in mathematics education

In this study, a quiz is defined as a set of questions that should be answered by students, and after submitting their answer, they will receive some feedback. The intention and implementation of quizzes may differ between instructors in different courses, and there are deviating results regarding the potential of quizzes on students' success on course results. In several studies, the potential of quizzes and assignments have been compared. In these studies, assignments refer to homework that students are free to choose

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when and how to work with, whereas quizzes refer to tests administered in a controlled setting wherein each student is solely responsible for their own work. In a comparison between graded assignments and graded quizzes in engineering courses, Fernandez *et al*. (2006) found that assignments did not have as much impact as quizzes on students' exam results, even though the assignments and quizzes contained the same types of problems. This was discussed to be potentially due to students relying on more knowledgeable peers' solutions for assignments and not placing much emphasis on understanding the content (Fernandez *et al*., 2006). Similarly, Gutarts & Bains (2010) did not find any impact at all from mandatory assignments on course results. However, they neither observed any significant difference between assignments (open setting) and weekly graded quizzes (administered in controlled settings), although there were indications that students who completed quizzes performed slightly better than those who completed assignments. On the contrary, Latif & Miles (2020) found significant positive effect of assignments on students' performance, while the introduction of quizzes did not have any positive effect overall. However, they could see some differences between groups of students, where international students' results benefited from quizzes while domestic students' results did not. At the same time, both Hannah *et al*. (2014) and Figueroa-Cañas & Sancho-Vinuesa (2021) showed a significant positive effect of optional quizzes on students' mathematical success. Similar findings are discussed by Gaspar Martins (2017) and Evans *et al*. (2021), where both grades and pass-rate increased in the courses with quizzes. In these studies, students had the opportunity to work on quizzes outside of class.

In several studies, doing quizzes could generate extra credits to the final grade, which is also discussed as an effective incitement for engagement (Gaspar Martins, 2017; Evans *et al*., 2021). A similar result is shown by Gutarts & Bains (2010), at least in terms of completing assignments, while there were indications that students completed quizzes despite not generating any credits. At the same time, Zerva *et al*. (2022) conclude that students opt out of optional quizzes that do not generate extra credits. Flexibility, quick responses and online format are features that are appreciated by students (Blanco & Ginovart, 2012; Rønning, 2017; Guo, 2022; Feudel & Unger, 2024). Particularly, flexibility and the possibility to multiple attempts are suggested to influence the time students actually spend working on the mathematical content and trying to solve the tasks and correct errors before completing a quiz (Guo, 2022). However, spending too much time on quizzes to achieve high scores is shown to have a negative effect on exam results (Hannah *et al*., 2014). It is furthermore discussed that students' study strategies depend on their study year, which in turn influence the effect quizzes may have on students' final results (Fernandez *et al*., 2006).

Results show that regular quizzes are perceived helpful by students for understanding the content (Gaspar Martins, 2017; Rønning, 2017; Evans *et al*., 2021; Fuchs & Sahmbi, 2025), and that quizzes promoted and increased students' engagement with the mathematical content (Blanco & Ginovart, 2012; Evans *et al*., 2021). On the contrary, Lowe (2015) showed that students' use of optional quizzes for regularly practising content was low in general, but increased around deadlines for summative assessments. In that study, the quizzes did not generate any credits to the final grade. At the same time, the students who actually used the quizzes found them helpful (Lowe, 2015). Different ways of using quizzes throughout a course aligns with results by Feudel & Unger (2024) who found six categories of use related to study strategies: rehearsal, elaboration, organization, planning, monitoring and regulation. They also found that students' use of quizzes depended on the type of tasks in the quizzes, to confirm knowledge or to gain deeper understanding. The type of tasks in a quiz depends on the purpose of the quiz and Rønning (2017) discusses that quizzes should not be used as high-stake assessments as this format does not seem to promote elaboration and thus is not beneficial for deep learning. On the other hand, the use of quizzes as low-stake assessments, for example for chapter checking, contributed to self-regulation and regular work (Blanco & Ginovart, 2012) as well as planning what to study more on

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(Fuchs & Sahmbi, 2025). Another way to use quizzes is as preparation for upcoming lectures for students to identify and prepare relevant questions for the teacher, which has proven beneficial for increased results (Fuchs & Sahmbi, 2025). It is also shown that quizzes as preparation increased students' engagement with the textbook in the course, and there were indications that this way of using quizzes can increase course results (Gyllen *et al*., 2021).

4. Method

The study is based on a quantitative approach, where various quantitative comparisons have been carried out to capture the possible influence of quizzes on students' mathematics results (RQ1). Furthermore, a survey has been sent out to collect data that can describe students' experiences of taking the quizzes (RQ2).

4.1. Participants

The participants were first year students enrolled in master-of-engineering programmes at one Swedish university. The academic year at the university is divided into four study periods September–November, November–January, January–March, and March–June. Each period encompasses nine to eleven weeks, of which the final week is the exam week. Two resit exams are offered within a year after the ordinary exam. There are no resit penalties applied to grades, and there is no limit to how many times a student may attempt any given exam, but once an exam is passed, students are not allowed to resist in order to obtain a higher grade. Students' scores on the regular exams for three different courses in the academic year 2023/2024 have been included in the analysis. The first course, linear algebra, was taught in the second study period. The second course, differential calculus, was taught in the third study period, and the third course, integral calculus, was taught in the fourth study period.

4.2. Quizzes

The purpose of the quizzes was to nudge students to engage in continuous learning with the content in the course. Not that the quizzes should be an additional curricular material beside the textbook, lectures, sessions for practising, and teaching materials on the learning platform. Nor, that the quizzes should cover all the content students need to study in the course, and replace recommended tasks from the textbook to practice on. At the same time, the idea was also that the immediate feedback in the form of whether it was a right or wrong answer, would help student find out areas/topics they needed to study more about. As such, the quizzes could be regarded as an additional study resource. The questions in the quizzes were constructed by the teacher to direct students to some of the main topics covered in the lectures that the teacher assessed as central. A range of different type of questions were used, such as: multiple-choice questions with one correct answer (Fig. 1) or several correct answers (Fig. 2); numerical questions where free form numerical answers were required (Fig. 3), and true/false questions (Fig. 4). In particular, using numerical questions instead of multiple-choice questions when it is possible reduces the opportunity to just try different alternatives until reaching the correct answer, and can instead directs students to read the textbook and/or lecture notes.

In total, 14 weekly quizzes were developed for the two consecutive courses, differential calculus and integral calculus, seven quizzes for each course with each quiz consisting of five to ten questions. Each quiz required approximately 2 to 3 hours to develop. There were no quizzes in the linear algebra course.

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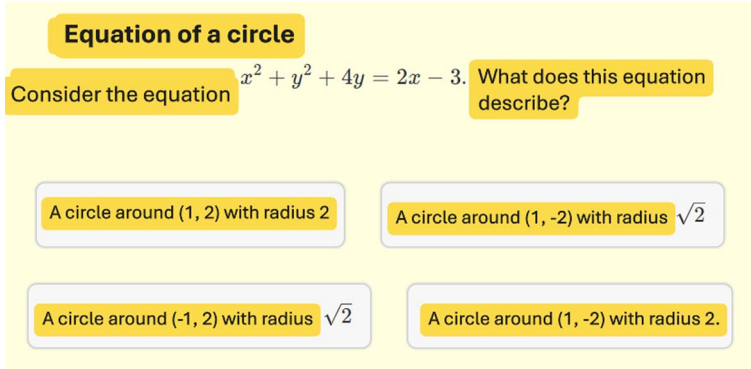

FIG. 1. Example of a multiple-choice question with one correct answer. Note. Translated from Swedish.

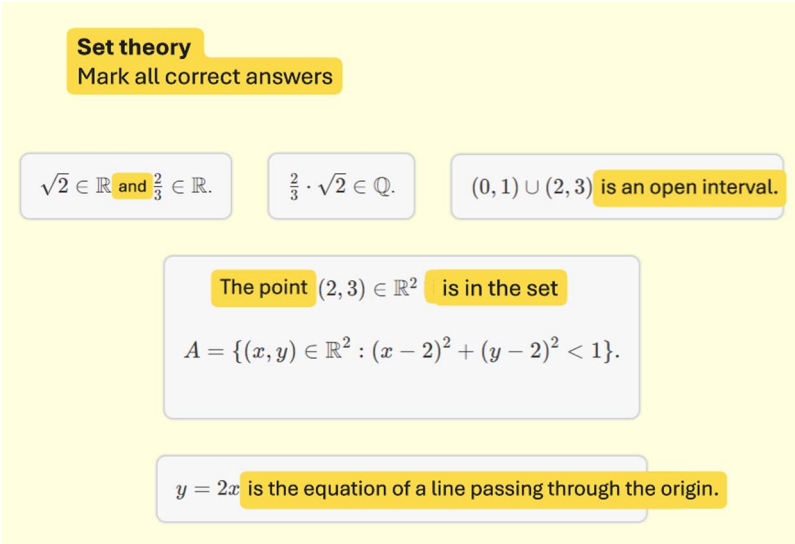

FIG. 2. Example of a multiple-choice question with multiple correct answers. Note. Translated from Swedish.

Educaora[1] was used as platform for implementing the quizzes, and links to the quizzes were provided through the learning platform Moodle. Students got access to the first quiz at noon on Wednesday in the first week of the course, after lectures had commenced, and the quiz closed at noon on Wednesday in the following week when the second quiz was opened for the students. This process continued in the same way for all quizzes in both courses. That is each quiz was only available to the students for one week, and questions in the quiz were related to the content in focus for the week the quiz opened. During this week, students had an unlimited number of attempts to solve each task in the quiz. This choice was based on the belief that the quizzes were thought to encourage continuous learning and provide a form of

[1] https://educaora.com/@educaora/Welcome

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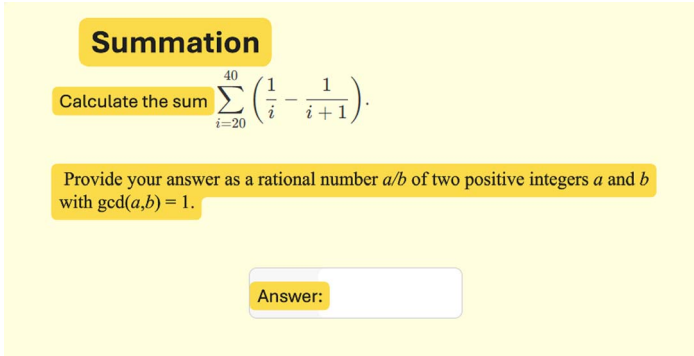

FIG. 3. Example of a numerical question. Note. Translated from Swedish.

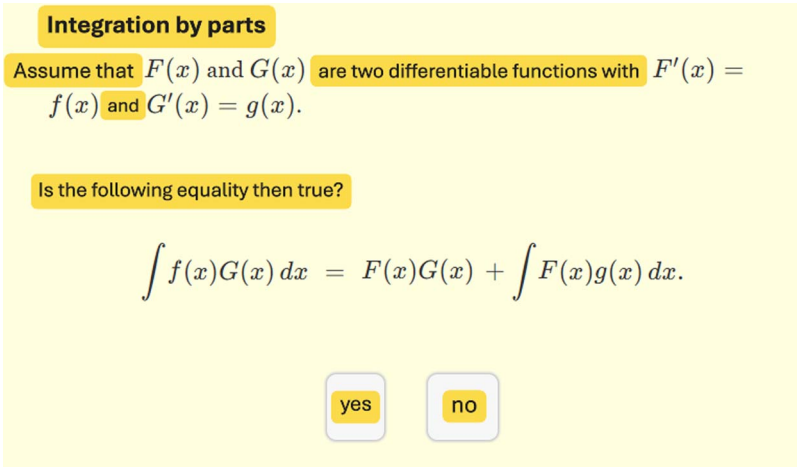

FIG. 4. Example of a true/false question. Note. Translated from Swedish.

early feedback, and not a form of knowledge control. All questions, as well as options in the questions, came in the same order for all students (i.e. there were no random distribution). Furthermore, a feature in Educaora is that students see only one question at a time and cannot proceed to the next question until they have answered the current one correctly. For multiple-choice questions, students may attempt the question repeatedly until they select the correct option; however, after each incorrect choice they must wait 120 seconds before they can make another selection. For questions that require entering a numerical value, students must also provide the correct value before moving on, but they can submit a new value immediately after an incorrect attempt (i.e. there is no waiting time). As an incentive for engaging in the quizzes, bonus points for the exam were awarded to the students if they finished enough quizzes in time. To receive two bonus points the students had to complete at least 80% of the quizzes, and to receive one bonus point at least 60% of the quizzes were required. Students also had a visit of one of the researchers in the first lecture of the Differential Calculus course to explain conditions for the quizzes and to inform about the purpose of the study.

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4.3. Data analysis with respect to RQ1

For each calculus course, we investigate how the number of bonus points from the quizzes affect the score on the exam. Since the individual scores on the exams also will be affected by each student's prior mathematical knowledge, we need to find a way to estimate each student's mathematical proficiency. For both calculus courses, we use the exam score on the regular exam on linear algebra as a measure of each student's mathematical proficiency. Comparison of students' exam results in these courses in 2019 shows a strong correlation between the results in linear algebra and the respective calculus course ($r = 0.72$ and $r = 0.62$, respectively), which supports the use of the exam results from linear algebra as a measure of mathematical proficiency.

For each student, included in the sample, a difference score is calculated, y_i, as the difference between the scores on the exams for the respective calculus course (excluding bonus points) and the linear algebra exam. The maximum score, excluding bonus points, is 24 points on the exams in all three courses.

Two student samples were selected: sample A for the differential calculus course, and sample B for the integral calculus course. The samples only included students who took the courses for the first time, i.e. students that were re-registered on these courses were not included in the samples. We also excluded students who scored zero points on either the linear algebra exam or the calculus exam from the samples, since it is unclear whether a serious attempt on that exam was made in those instances. Sample A, thus consists of first-year students who took the regular exams (and scored points) on both linear algebra and differential calculus, and sample B similarly consist of first-year students who took the regular exams (and scored points) on both linear algebra and integral calculus, 50 students are included in sample A, and 47 students in sample B. Student results on exams are publicly available information in Sweden; thus, no consent from students is required for these kinds of analysis.

In the data analysis student i has x_i bonus points and a difference score y_i. Both x and y are discrete variables. We expect a monotonic relationship between x and y, but not necessarily a linear relationship. Fractional ranks $R\left(x_i\right)$ and $R\left(y_i\right)$ are calculated. Spearman's rank correlation coefficients, r_s, are then calculated as $r_s = \operatorname{cov}\left[R\left(x_i\right), R\left(y_i\right)\right] / \left[\sigma_{R(x_i)} \bullet \sigma_{R(y_i)}\right]$. The null hypothesis is that the correlation is zero.

4.4. Student surveys (RQ2)

Two online surveys were distributed to all students registered on the differential calculus course ($n = 117$). One in the middle of the course, after the third quiz, and one after the regular exam. Furthermore, a third online survey was distributed to all students registered on the Integral Calculus course ($n = 112$) the week before the regular exam of that course. These samples include not only the first-year students (as for the samples used for RQ1), but also re-registered students. In accordance with ethical guidelines (Swedish Research Council, 2024), the start page of the survey included information about the purpose of the study, that the survey was voluntary, and that no personal information was collected (i.e. an anonymous survey). To maximize response rate, students were given time in a lecture to complete each survey. Additionally, students that participated in the first lecture of the differential calculus course also had a visit of one of the researchers and got oral information about the coming surveys and their purpose. The response rate for the first two surveys were 43% ($n = 50$) and 42% ($n = 49$), respectively, and 26% ($n = 29$) for the third survey. Due to the setup, it is reasonable to assume that students who are regularly attending lectures may be overrepresented among the respondents.

Items in the surveys were based on questions used in previous studies related to students' experiences about quizzes, as well as identified perceptions (e.g. Blanco & Ginovart, 2012; Gaspar Martins, 2017). Additionally, some items were created to capture specifics for the context of this study. The first survey

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TABLE 1. *Average scores for all students scoring on the regular exam for the three different courses and for those students included in Sample A and Sample B. Bonus points are not included in the averages*

Course	# Registered students	# Students scoring on the exam	Average score		
			All (SD)	Sample A (SD)	Sample B (SD)
Linear algebra	82	60	13.1 (6.2)	13.4 (5.9)	14.6 (5.7)
Differential calculus	77	56	13.1 (5.0)	13.4 (5.1)	
Integral calculus	73	51	11.8 (4.2)		12.1 (4.0)

Note. Standard deviations (SD) are given in parentheses.

included six items, the second survey included nine items, and the third survey included eight items. In the first survey, one item included seven statements about the benefits of the quizzes, and students were asked to choose whether they agreed or not. Examples of statements: 'Quizzes remind me to study the subject every week', 'I learn new mathematics answering to quizzes', and 'I do not bother doing the quizzes'. Some of the items reoccurred in two or three of the surveys. Examples of same items in all three surveys were: 'How many of the quizzes have you solved?' and 'In how many of the lectures have you participated?'. To capture students' willingness to complete the quizzes, students that answered that they completed most of the quizzes in both survey 2 and 3 had to decide if they would have studied less/equal/more without the quizzes, and if they would have solved the quizzes without the bonus points. All students in both surveys also got to rate how useful for learning they experienced the quizzes. Furthermore, survey 2 and 3 included an item where students were asked to rank how useful (from most to least) they had found the different provided learning resources they used in the courses, including the quizzes. See Appendix A for a full list of items on the surveys, including options for answers. All surveys also ended with a possibility for students to add anything else they wanted about their experiences of the quizzes. Descriptive analysis of the responses is presented in results.

5. Results

First, some general results about student participation are presented, and then the results for each research question are presented.

5.1. General information about the participation in the courses

From Table 1, we see that the number of registered first-year students gradually decreases during the academic year. There are five students less on differential calculus than on linear algebra. None of the five students that dropped out took the regular exam on the linear algebra course. An additional four students dropped out prior to the integral calculus course. Three of those students had not written any of the two previous regular exams, and the fourth student had failed on both prior exams. Hence, eight out of the nine students that dropped out did not write any of the regular exams. In total, there were 15 of the 82 students registered on linear algebra who did not write any of the three regular exams.

Of the 82 first-year students registered on the linear algebra course, 21 students did not attend the regular exam, one attended but score zero points, which resulted in 60 students that attended the regular exam and scored more than zero points. Similarly, 56 and 51 students scored on the exams on differential calculus and integral calculus, respectively, (Table 1).

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TABLE 2. *Average scores for students who had 0 or 1 bonus point and 2 bonus points, respectively, on differential calculus (Sample A) and on integral calculus (Sample B). Bonus points are not included in the averages*

Sample	Bonus points	#	Average score			Average difference score (SD)
			Linear algebra (SD)	Differential calculus (SD)	Integral calculus (SD)	
A	0–1	11	12.0 (6.9)	9.8 (4.9)		−2.1 (5.7)
	2	39	13.8 (5.7)	14.4 (4.7)		0.5 (4.4)
B	0–1	19	15.5 (6.6)		11.3 (4.5)	−4.2 (5.1)
	2	28	14.1 (5.7)		12.6 (3.6)	−1.5 (3.8)

Note. Included is also the average difference score for each group of students. SD are given in parentheses.

TABLE 3. *Spearman rank correlation coefficients, r_s-, and P-values calculated from the difference scores for students who wrote both the exam on linear algebra and on the respective calculus course*

Course	Sample size (N)	r_s	P
Differential calculus (Sample A)	50	0.160	0.267
Integral calculus (Sample B)	47	0.325	0.026*

*Significance level $P < 0.05$.

Comparing average scores for the different cohorts show that students included in Sample A or B have higher scores than those students who have not taken one of the two exams included in respective sample. For example, average score for all 60 students taken the linear algebra exam was 13.1 and 14.6 for those included in sample B (Table 1, last column). This implies that average score for the 13 students that did not score on the integral calculus course was 7.8.

5.2. *Effect of the quizzes (RQ1)*

Table 2 shows that for Sample A (differential calculus), students who obtained two bonus points from quizzes scored higher on both linear algebra and differential calculus compared to those with 0 or 1 bonus point. However, the difference between the groups was larger in differential calculus than in linear algebra, suggesting that the quizzes contributed to improved performance in the calculus course beyond what could be expected from prior proficiency. For Sample B (integral calculus), the pattern was somewhat different. Students with two bonus points had slightly lower average scores on linear algebra compared to those with 0 or 1 bonus point, yet they outperformed the latter group on the integral calculus exam. This indicates that the quizzes may have helped compensate for initially lower proficiency and supported achievement in the calculus course. Taken together, in both samples the difference scores were more favourable for students who obtained two bonus points, which suggests a positive effect of regular quiz participation on exam performance. Note that since there were relatively few students with 0 bonus points, and even fewer with 1 bonus point, we have chosen for statistical reasons to only divide the students into two groups in Table 2, those with 0 or 1 bonus point constituting one group, and those with two bonus points constituting the other group.

In Table 3, it is also seen that the correlation coefficients are positive for both samples, although only significant on a 5% level for Sample B.

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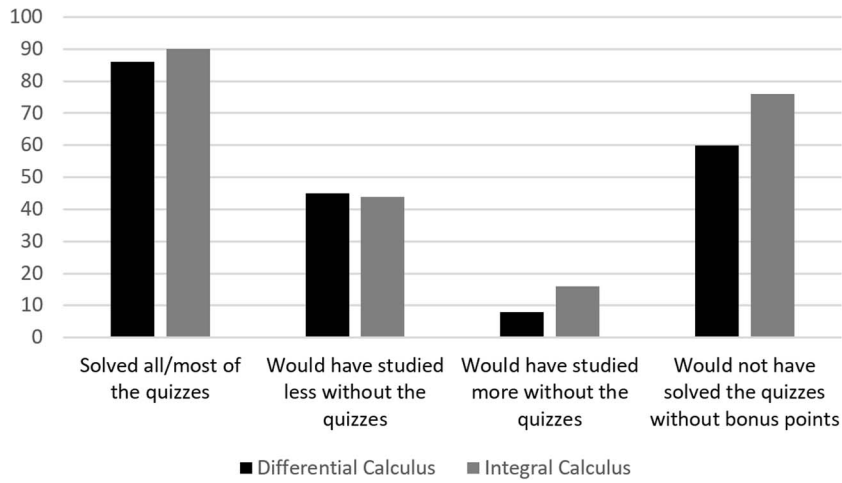

FIG. 5. Percentage of responding students in each course who solved all or most quizzes, and conditional percentages (based on this group) regarding perceived impact on study time and the role of bonus points. Note: The columns represent those who answered 'yes' on respective question, and the last three questions were only posed to those who answered 'yes' on the first question.

TABLE 4. *Proportion of students agreeing to statements in survey 1 about quizzes' contribution to learning mathematics*

Statement	Yes (%)
Quizzes remind me to study the subject every week	82
Quizzes help me realize there are parts of the content I thought I knew but I did not	86
Quizzes help me to have a better perception of the level I am reaching	80
I learn new mathematics by answering the quizzes	70

5.3. Students experience of the quizzes (RQ2)

Of the 48 students that responded to survey 2 in the differential calculus course, 40 reported that they had participated in all or most of the given lectures. On survey 3 in the integral calculus course 21 of 27 respondents reported that they had participated in all or most of the given lectures. A majority of responding students stated that they had solved all or most of the quizzes in both courses, 86 and 90%, respectively (Fig. 5). Of these, 45 and 44%, respectively, estimated that they would have studied less without the quizzes, while 8 and 16%, respectively, estimated that they would have spent more time studying if they had not solved the quizzes.

Additionally, 60 and 76%, respectively, of these students would not have solved the quizzes without the bonus points (Fig. 5). Given that students had time to fill out the survey during a lecture, it is reasonable to assume that most students taking the quizzes also were the ones that participating in most of the lectures.

Furthermore, a majority of all responding students in survey 2 and 3 (71 and 82%, respectively) experienced the quizzes as useful for learning. Table 4 shows students' experiences related to statements about specific possible learning contributions of the quizzes. As a comparison, 10% of the students in survey 1 answered that they did not put much effort in solving the quizzes.

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TABLE 5. *RM and AM for the different learning resources in both courses*

	RM (%)		AM (%)	
Learning resource	Survey 2	Survey 3	Survey 2	Survey 3
Solved exams	86	93	74	71
Textbook	84	86	56	41
Lectures	82	90	86	75
Quizzes	78	79	46	53
Lecture notes	76	76	40	38
Practice sessions	69	83	24	19
Recorded lectures	69	†	21	†
Solved examples	59	79	46	50

Note. Learning resources are ordered by RM from survey 2.
†There were no recorded lectures in the Integral Calculus course.

To compare students' ranking of quizzes in relation to the different learning resources they had used in the course, a rank measure (RM) and an appreciation measure (AM) were calculated for each learning resource. RM is obtained by dividing the number of students who ranked a particular learning resource by the total number of students who answered the survey item. For example, 38 students ranked quizzes (i.e. indicated that they had used this resource), and 49 students in total responded to the rank-item in survey 2, thus $RM_{Quizzes} = 38/49$ (Table 5, row 4, column 2). AM is obtained through pairwise comparing ranking of a particular learning resource with the other ranked learning resources. This is done by dividing the number of times a specific learning resource was pairwise ranked higher by the total number of times the specific learning resource was pairwise ranked. For example, in survey 3, quizzes were ranked higher than lectures 17 times, ranked higher than the textbook 12 times, and so on. In total, quizzes were pairwise ranked higher 71 times compared to all other learning resources, and pairwise ranked 134 times, thus $AM_{Quizzes} = 71/134$ (Table 5, row 4, column 5).

As seen in Table 5 (row 1), solved exams have the highest RM for both courses and the second highest AM. This implies that almost all students that responded to the survey have used the solved exams as a learning resource, and most of them considered solved exams to be one of the most useful learning resources in the course. The highest ranked learning resource in both courses is the lectures (AM 86 and 75%, respectively), which also have a high RM (i.e. many students have ranked them). This indicates that many of those students who responded to the surveys also were the ones attending the lectures.

RM for the quizzes (Table 5, row 4) implies that about 80% of the students who answered the survey had used the quizzes, or at least some of them. AM for the quizzes shows that they are the third highest ranked learning resource in the Integral Calculus course, and the fourth highest ranked in the differential calculus course (together with solved examples).

6. Discussion and conclusion

The present study investigated the potential of weekly online quizzes to support first-year engineering students in succeeding in university mathematics courses. The findings provide empirical support that quizzes, when used regularly and in a low-stakes format, can contribute positively to students' learning outcomes. The findings provide partial but noteworthy answers to both research questions. With respect to RQ1, the analysis showed a positive, albeit moderate, association between quiz participation and exam performance, most clearly visible in the integral calculus course where a statistically significant correlation was found. Although the effect sizes were not large, they nevertheless suggest that quizzes

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may contribute to students' understanding and retention of mathematical content, particularly when implemented regularly and in a low-stakes format. These findings align with previous studies highlighting the potential of formative assessment such as quizzes to be effective in promoting engagement and learning, particularly when coupled with immediate feedback and flexible access (e.g. Gaspar Martins, 2017; Evans *et al*., 2021; Figueroa-Cañas & Sancho-Vinuesa, 2021). However, as with other quasi-experimental designs, these associations must be interpreted with caution, particularly in the absence of a control group and with the acknowledged limitations in the measurement of prior knowledge.

With respect to RQ2, survey responses indicated that the majority of students perceived the quizzes as beneficial to their learning. Most notably, students reported that the quizzes helped them stay on track with weekly studies and identify gaps in their understanding, features commonly associated with formative assessment. Nevertheless, quizzes were not rated as the most useful learning resource. In both courses, lectures were consistently ranked higher in terms of usefulness, suggesting that students continue to view teacher-led instruction as central to their learning experience, and regard quizzes more as a complementary tool. This resonates with earlier findings that formative quizzes can promote regular engagement but cannot substitute for comprehensive instruction (Rønning, 2017; Fuchs & Sahmbi, 2025).

An important finding is that the quizzes primarily reached students who are already active and engaged in the course. Most students who completed the quizzes are most likely also the ones that reported regular attendance at lectures, indicating that the quizzes may not reach or engage students who are less engaged or at greater risk of failure. This finding is consistent with Lowe (2015), who observed that students' use of optional quizzes was limited outside of high-stakes assessment periods. Thus, while quizzes may help reinforce habits of regular learning among already committed students, they are unlikely to suffice as a stand-alone intervention for improving pass rates in engineering mathematics more broadly.

Moreover, the incentive structure appears to have been critical in engaging student participation. A majority of students indicated they would not have completed the quizzes in the absence of bonus points towards the exam. This supports earlier findings that grade incentives can effectively prompt engagement with formative tasks (e.g. Gaspar Martins, 2017; Evans *et al*., 2021), and that a lack of this type of incentive may reduce student engagement (Zerva *et al*., 2022). However, such reliance on external rewards raises questions about the development of engagement and sustainable self-regulated learning behaviours. Future studies should therefore explore how quizzes might be integrated in ways that foster engagement, for example by linking quizzes more explicitly to preparatory work for lectures or problem-solving sessions. At the same time, the findings suggest that quizzes alone cannot resolve the issue of low throughput in engineering mathematics. As Johansson *et al*. (2021) have emphasized, regular engagement with content is essential for student success, but promoting such engagement, particularly among less active students, requires more comprehensive and multifaceted strategies. Büchele (2023) shows that students who predominantly adopt a memorization-oriented approach are less likely to engage in regular learning practices. Seen in this light, quiz participation largely confined to students who are already active in the course may reflect differences in underlying learning strategies. While quizzes appear to support students who are already inclined towards regular engagement, they may be less effective for students whose study practices are characterized by surface-level memorization and episodic exam preparation. This suggests that quizzes, although beneficial, are unlikely to shift learning behaviours on their own. To reach less engaged students and improve overall progression, quiz-based interventions may need to be complemented by explicit pedagogical efforts aimed at addressing students' conceptions of learning mathematics and encouraging deeper, more conceptually oriented study strategies. Quizzes may thus be best understood as one useful element in a broader set of interventions aimed at supporting continuous learning.

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Several limitations of the present study must be acknowledged. The design was quasi-experimental, with no randomized control group, which limits the causal inferences that can be drawn. Furthermore, survey respondents were most likely overrepresented by students who regularly attended lectures, leading to possible bias in the perceptions reported. It cannot be ruled out that students who performed better on the exams would also have done so in the absence of quizzes, due to their overall study habits, motivation, or engagement. From this perspective, the results should be interpreted as indicating a relationship rather than an effect in a strict causal sense. The quizzes may function more as a supportive structure that reinforces existing patterns of regular learning, rather than as an intervention that fundamentally alters students' engagement or study strategies. At the same time, the surveys show that approximately 70–80% of responding students experienced the quizzes as useful for their learning, and a substantial proportion reported that they would have studied less without the quizzes. While such self-reports cannot establish causal effects on performance, they nevertheless suggest that students perceive the quizzes as adding value to their learning process, particularly by prompting regular study and helping them identify gaps in understanding. Taken together, the findings therefore support a cautious interpretation in which quizzes are viewed as a beneficial component of the learning environment for engaged students, rather than as a stand-alone mechanism for improving outcomes across the entire student cohort.

Another limitation is that the analyses relied on using exam scores in linear algebra as a proxy for students' prior mathematical proficiency, a measure that may not fully capture differences in background knowledge. Additionally, the range of possible difference scores depends on the students' score on the linear algebra exam, where the range of possible difference scores is skewed towards lower values the higher the student's score on the linear algebra exam is. For example, a student who scored a maximum of 24 points on the linear algebra exam cannot get a higher difference score than zero. This truncation problem affects the accuracy of the calculated r_s- and P-values. Tests that remove students with very low and very high scores on the linear algebra exam from samples A and B shows that these values are very sensitive to how the samples are chosen. Hence, the uncertainties in the presented r_s- and P-values are quite large, and the difference in the values for the two courses in Table 3 can be partly attributed to the truncation problem.

Future research should therefore examine not only the immediate effects of quizzes on exam performance but also their long-term influence on student retention, study strategies and engagement. Qualitative investigations of how students actually use quizzes, whether as rehearsal, elaboration or regulation (cf. Feudel & Unger, 2024), could provide deeper insights into how quizzes support or hinder meaningful learning. Designing interventions that reach less engaged students, and that combine quizzes with other supportive measures, remains an important task for further work.

In conclusion, weekly online quizzes cannot in themselves solve the challenge of low pass rates in engineering mathematics, but they can play a meaningful role in scaffolding students' study habits and providing timely feedback. Their potential lies not in replacing existing teaching formats but in complementing them, helping committed students to engage regularly with mathematical content. The key challenge ahead is to develop strategies that extend such benefits to those students who are most at risk of disengagement.

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Appendix A. All survey items are translated from Swedish

Survey 1

(1) Seven different statements about the quizzes. Choose 'yes' or 'no':

  (a) Quizzes remind me to study the subject every week.
  (b) Quizzes help me realize there are parts of the content I thought I knew but I did not.
  (c) Quizzes help me to have a better perception of the level I am reaching.
  (d) I learn new mathematics by answering the quizzes.
  (e) I do not put much effort into solving the quizzes, I just 'try out' or copy the results (from peers, text book, web, etc.).
  (f) I do not bother doing the quizzes.
  (g) The quizzes stress me.

(2) How many of the quizzes have you solved so far? Choose one option: all; most; few; none
(3) How do you usually solve tasks in the quizzes? Choose the most appropriate option: alone; alone/together with someone; together with someone; copy and/or try out; not at all
(4) In how many of the lectures have you participated? Choose one option: all; most; half; few/none
(5) In how many of the scheduled 'training sessions' have you participated? Choose one option: all/most; half; few/none
(6) Seven different statements about recorded lectures. Do you agree? Choose 'yes' or 'no':

  (a) I watch the recordings when they are released
  (b) I use the recordings to help solve the quizzes
  (c) The recordings help me understand the content discussed during the lectures
  (d) I watch the recordings instead of participating in the lectures
  (e) I have watched a few of the recordings
  (f) I have no need to watch the recordings

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Survey 2

(1) How many of the quizzes have you solved during the course? Choose one option: all; most; few; none

If answering 'all' or 'most,' they also got to answer item 2) and 3)

(2) Without the quizzes, I would have studied: Choose the most appropriate option: less; equal; more
(3) I would have solved the quizzes without the bonus points? Answer: yes or no

Rest of the items for all students

(4) Without the quizzes, I think I would have scored ... on the exam. Choose the most appropriate option: better, the same; worse
(5) I found the quizzes to be .... Choose the most appropriate option: useful for learning; indifferent for learning; an obstacle for learning
(6) In how many of the lectures have you participated? Choose one option: all; most; half; few/none
(7) In how many of the scheduled 'training sessions' have you participated? Choose one option: all/most; half; few/none
(8) Eight different statements about recorded lectures. Do you agree? Choose 'yes' or 'no':

(a) I watch the recordings when they are released
(b) I use the recordings to help solve the quizzes
(c) The recordings help me understand the content discussed during the lectures
(d) I watch the recordings instead of participating in the lectures
(e) I have watched the recordings as revision for the exam
(f) I have watched a few of the recordings
(g) I haven't watched any of the recordings

(9) Rank the following resources according to their helpfulness for you to learn the content in the course and prepare for the exam. Only rank the ones you used, from most to least helpful

(a) Lectures
(b) Recorded lectures
(c) Provided lecture notes (pdf)
(d) Scheduled 'training sessions'
(e) Provided solutions to recommended tasks (pdf)
(f) Quizzes
(g) The text book
(h) Solved exams (pdf)

Survey 3

(1) How many of the quizzes have you solved during the course? Choose one option: all; most; few; none

If answering 'all' or 'most,' they also got to answer item 2) and 3)

(2) Without the quizzes, I would have studied: Choose the most appropriate option: less; equal; more
(3) I would have solved the quizzes without the bonus points? Answer: yes or no

Rest of the items for all students

(4) I found the quizzes to be .... Choose the most appropriate option: useful for learning; indifferent for learning; an obstacle for learning

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(5) How did you usually solve tasks in the quizzes? Choose the most appropriate option: alone; alone/together with someone; together with someone; copy and/or try out; not at all
(6) In how many of the lectures have you participated? Choose one option: all; most; half; few/none
(7) In how many of the scheduled 'training sessions' have you participated? Choose one option: all/most; half; few/none
(8) Rank the following resources according to their helpfulness for you to learn the content in the course and prepare for the exam. Only rank the ones you used, from most to least helpful

    (a) Lectures
    (b) Provided lecture notes (pdf)
    (c) Scheduled 'training sessions'
    (d) Provided solutions to recommended tasks (pdf)
    (e) Quizzes
    (f) The text book

Solved exams (pdf)

Helena Johansson is an associate professor of mathematics education at Mid Sweden University. Her main research interests concern the use of mathematical symbols and how various uses may influence understanding, primarily in mathematics but also in physics, as well as how different designs of mathematics tasks may influence students' mathematical understanding. Her research is guided by theories related to mathematical knowledge, mathematical struggles and conceptual development, cohesion and tension between mathematical representations, and the objectification of mathematics. Her research spans mathematics education from grade 1 to university studies.

Peter Glans is an associate professor of physics at Mid Sweden University. He received his PhD from Uppsala University in 1994. His previous research is mainly in atomic and molecular physics. He also has an interest in science education research and many years of experience in both teaching and as program coordinator for engineering students.

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